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Unit – 5 : Session – 6 : SLO – 2 | Minimum Spanning Tree — Kruskal's Algorithm

■ I. Graph Reference

- Edges with weights: A-B(10), A-C(12), B-C(9), B-D(8), C-E(3), C-F(1), D-E(7), E-F(3),
- D-G(8), D-H(5), F-H(6), G-H(9), G-I(2), H-I(11).

■ II. Fill in the Blanks

- Kruskal's algorithm is a GREEDY based technique.
- In Kruskal's algorithm, all edges are sorted in ASCENDING (increasing) order of their weights.
- The MST formed by Kruskal's algorithm always contains $n-1$ edges for a graph with n vertices.
- To avoid cycles in Kruskal's algorithm, we use UNION-FIND (disjoint set) to check if two vertices are in the same set.
- The time complexity of Kruskal's algorithm using Union-Find is $O(E \log E)$ (i.e. $O(E \log V)$).

■ III. True or False

- Kruskal's algorithm works only for connected graphs — TRUE (for a spanning TREE; on disconnected graphs it yields a minimum spanning forest).
- Kruskal's algorithm starts from an arbitrary node — FALSE (it starts from the globally lightest edge).
- Kruskal's algorithm always selects the lightest edge that does not form a cycle — TRUE.
- MST by Kruskal's and Prim's algorithms will always be identical for any graph — FALSE
- (total weight is the same, but the edge set may differ when equal-weight edges exist).

■ IV. Short Answer Questions

- Union-Find: maintains disjoint sets of vertices; FIND checks whether the two endpoints of an edge are already
- in the same set (which would form a cycle), and UNION merges the two sets when an edge is accepted.
- Kruskal's skips an edge when both its endpoints already belong to the same set, because adding it would
- create a cycle.
- Edges are sorted by weight so that at every step the globally lightest remaining edge is considered,
- guaranteeing the greedy choice is optimal (cut property) and yielding a minimum total weight.

■ V. Apply Kruskal's Algorithm

- Sorted edges: C-F(1), G-I(2), C-E(3), E-F(3), D-H(5), F-H(6), D-E(7), B-D(8), D-G(8), B-C(9), G-H(9), A-B(10), H-I(11), A-C(12).
- Step 1: C-F (1) -> accept. Sets: {C,F}
- Step 2: G-I (2) -> accept. Sets: {C,F}, {G,I}
- Step 3: C-E (3) -> accept. Sets: {C,F,E}, {G,I}
- Step 4: E-F (3) -> SKIP (E and F in same set; would form a cycle).

- Step 5: D-H (5) -> accept. Sets: {C,F,E}, {G,I}, {D,H}
- Step 6: F-H (6) -> accept (merges). Sets: {C,F,E,D,H}, {G,I}
- Step 7: D-E (7) -> SKIP (same set).
- Step 8: B-D (8) -> accept. Sets: {B,C,D,E,F,H}, {G,I}
- Step 9: D-G (8) -> accept (merges). Set: {B,C,D,E,F,G,H,I}
- Step 10: B-C(9), G-H(9) -> SKIP (cycles).
- Step 11: A-B (10) -> accept. All 9 vertices connected; 8 edges selected. STOP.
- Order of edges selected: C-F(1), G-I(2), C-E(3), D-H(5), F-H(6), B-D(8), D-G(8), A-B(10).
- TOTAL WEIGHT OF MST = $1 + 2 + 3 + 5 + 6 + 8 + 8 + 10 = 43$.